Transformations & Symmetry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

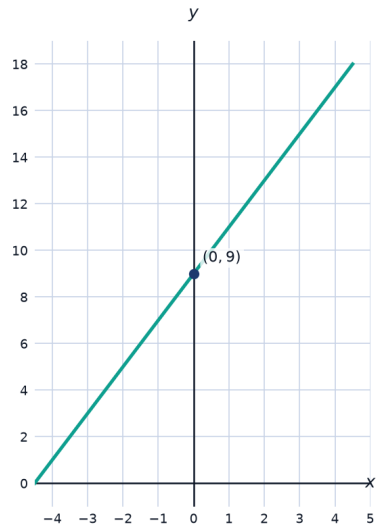
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC studying properties of rectangles, rhombuses, and squares, $m\angle A = 34^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(2)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for properties of trapezoids consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Transformations & Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

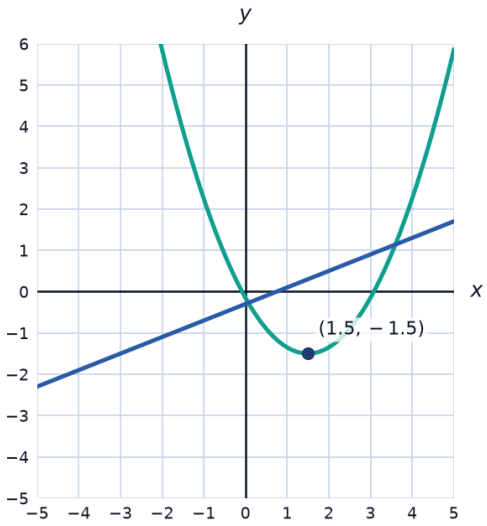
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the magnitude of $\mathbf{v} = \begin{pmatrix} -5 \\ 12 \end{pmatrix}$ and hence find a unit vector in the direction of $\mathbf{v}$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A price increased from 80 to 100 SAR. Find the percentage increase.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the linear system $3x + 2y = 16$ and $x - y = 2$ for x and y.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A car travels a distance of $d = 180$ km correct to the nearest 10 km in a time of $t = 2.4$ hours correct to 1 decimal place. Calculate the upper bound for the average speed of the car in km/h.

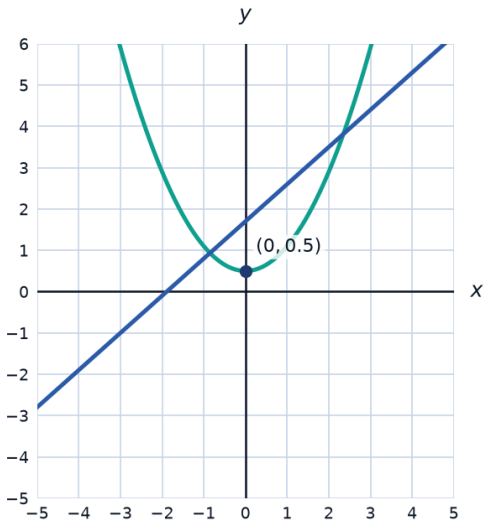
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedA researcher randomly assigns 50 students to a tutoring programme and 50 to a control group, then compares test scores. Which conclusion is best supported if the tutored group scores significantly higher? (A) Tutoring causes higher scores for the population sampled. (B) There is an association, but causation cannot be determined. (C) Results apply to all high school students nationwide. (D) The sample size is too small to draw any valid conclusion.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A USD 80 item is marked down 20%. Find the sale price.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The equation of a circle is given by $x^2 + y^2 + 10x - 4y + 20 = 0$. What is the radius of this circle?

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

If 5 workers finish a task in 12 days, how many days for 3 workers?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A shop offers a 15% discount on a 200 SAR item. Find the sale price.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 4 · SynthesisIn the system of equations $ax + 8y = 24$ and $6x + by = 8$, a and b are constants. If the system has infinitely many solutions, what is the value of $a + b$?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

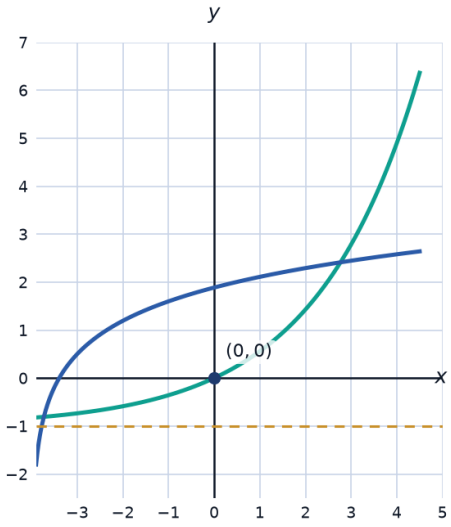
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Triangles & Congruence

Level 4 · SynthesisLine ℓ passes through $(2, -1)$ and is perpendicular to the line $3x + 6y = 18$. If line ℓ also passes through $(k, 9)$, what is the value of k?

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Solve the logarithmic equation $\log_3(x^2 - 7) = 2$ for all real solutions, and verify that all roots lie within the valid domain.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A line is parallel to $y = 2x + 1$. What is its slope?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

Two angles of a triangle are 55° and 75°. Find the third.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Circles

A car travels 120 miles in 3 hours. Find the average speed.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Area, Surface Area & Volume

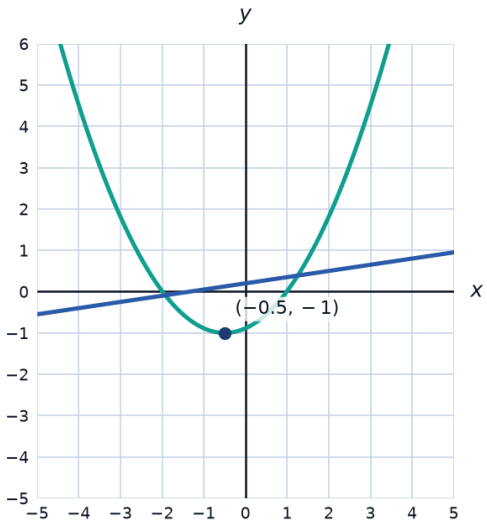
Point $M(3, -2)$ is the midpoint of line segment $\overline{AB}$. If point A has coordinates $(-1, 4)$, what are the coordinates of point B?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 103°

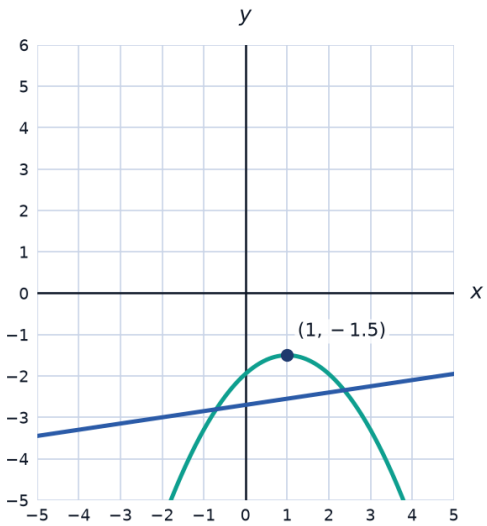
b. 113°

c. 95°

d. 148°

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying properties of parallelograms, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 102°

b. 112°

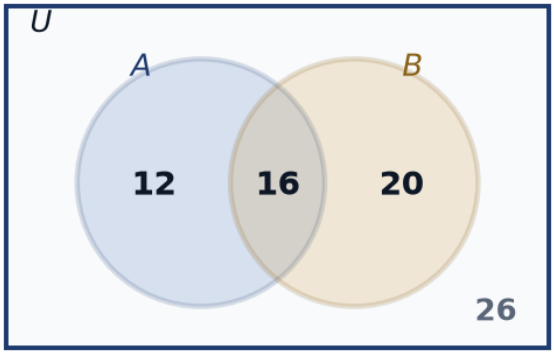
c. 94°

d. 147°

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for classifying quadrilaterals consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?



a. 16.5

b. 17.5

c. 18.5

d. 20.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 3$ in classifying quadrilaterals, evaluate $f(4)$.

a. 15

b. 18

c. 12

d. 19

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for coordinate methods with quadrilaterals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 4x + 4$ in coordinate methods with quadrilaterals, evaluate $f(4)$.

a. 20

b. 24

c. 16

d. 24

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Under a translation by the vector $\langle 5, -3 \rangle$, the image of point P is $P'(2, 1)$. What are the coordinates of the pre-image point P?

a. $(-3, 4)$

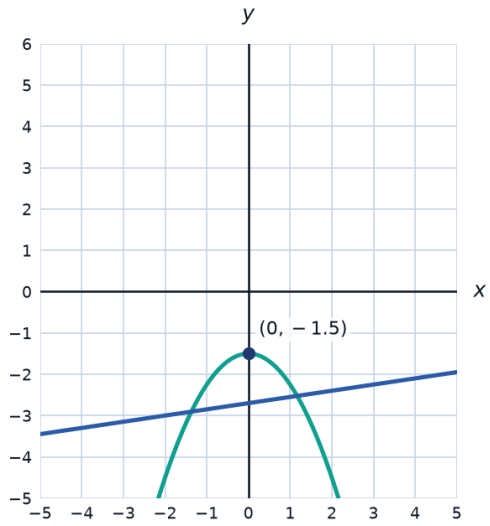
b. $(7, -2)$

c. $(-3, -2)$

d. $(7, 4)$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the order of rotational symmetry of a regular decagon?



a. 20

b. 9

c. 10

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the order of rotational symmetry of a regular dodecagon?

a. 1

b. 24

c. 11

d. 12

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the order of rotational symmetry of an equilateral triangle?

a. 1

b. 2

c. 3

d. 6

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the order of rotational symmetry of a square?

a. 8

b. 1

c. 4

d. 3

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the order of rotational symmetry of a regular pentagon?

a. 4

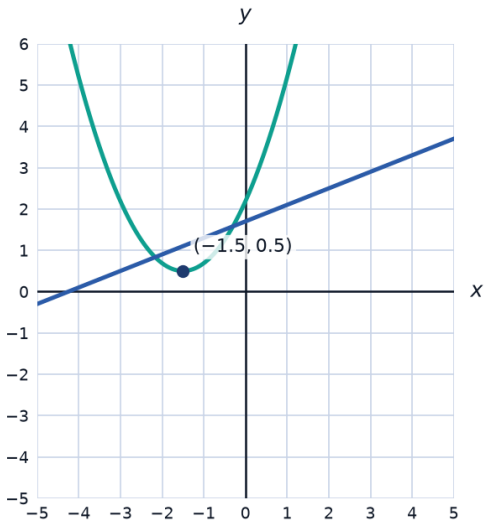
b. 5

c. 1

d. 10

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What is the order of rotational symmetry of a regular hexagon?



a. 5

b. 12

c. 6

d. 1

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the order of rotational symmetry of a regular heptagon?

a. 6

b. 7

c. 1

d. 14

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the order of rotational symmetry of a regular octagon?

a. 8

b. 7

c. 16

d. 1

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the order of rotational symmetry of a regular nonagon?

a. 18

b. 1

c. 8

d. 9

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Reflecting $(-2, 3)$ over the line $y = x$ gives:

a. (2,-3)

b. (3,-2)

c. (-2,3)

d. (-3,2)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

Reflecting $(3, -2)$ over x-axis gives:

a. (-2,3)

b. (-3,2)

c. (3,2)

d. (3,-2)

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Circles

Reflecting $(4, 5)$ over y-axis gives:

a. (-4,-5)

b. (4,5)

c. (5,4)

d. (-4,5)

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Area, Surface Area & Volume

Reflect $(3, -2)$ over the x-axis, then translate by $\langle 1, 2 \rangle$. Find the final image.

a. (4,4)

b. (4,0)

c. (-2,0)

d. (3,2)

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Foundations & Reasoning

How many lines of symmetry does a regular decagon have?

a. 9

b. 20

c. 10

d. 11

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Coordinate Geometry

How many lines of symmetry does a regular dodecagon have?

a. 11

b. 12

c. 13

d. 24

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Probability

How many lines of symmetry does an equilateral triangle have?

a. 2

b. 6

c. 3

d. 4

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Lines & Angles

How many lines of symmetry does a square have?

a. 8

b. 3

c. 4

d. 5

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Triangles & Congruence

How many lines of symmetry does a regular pentagon have?

a. 10

b. 5

c. 4

d. 6

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